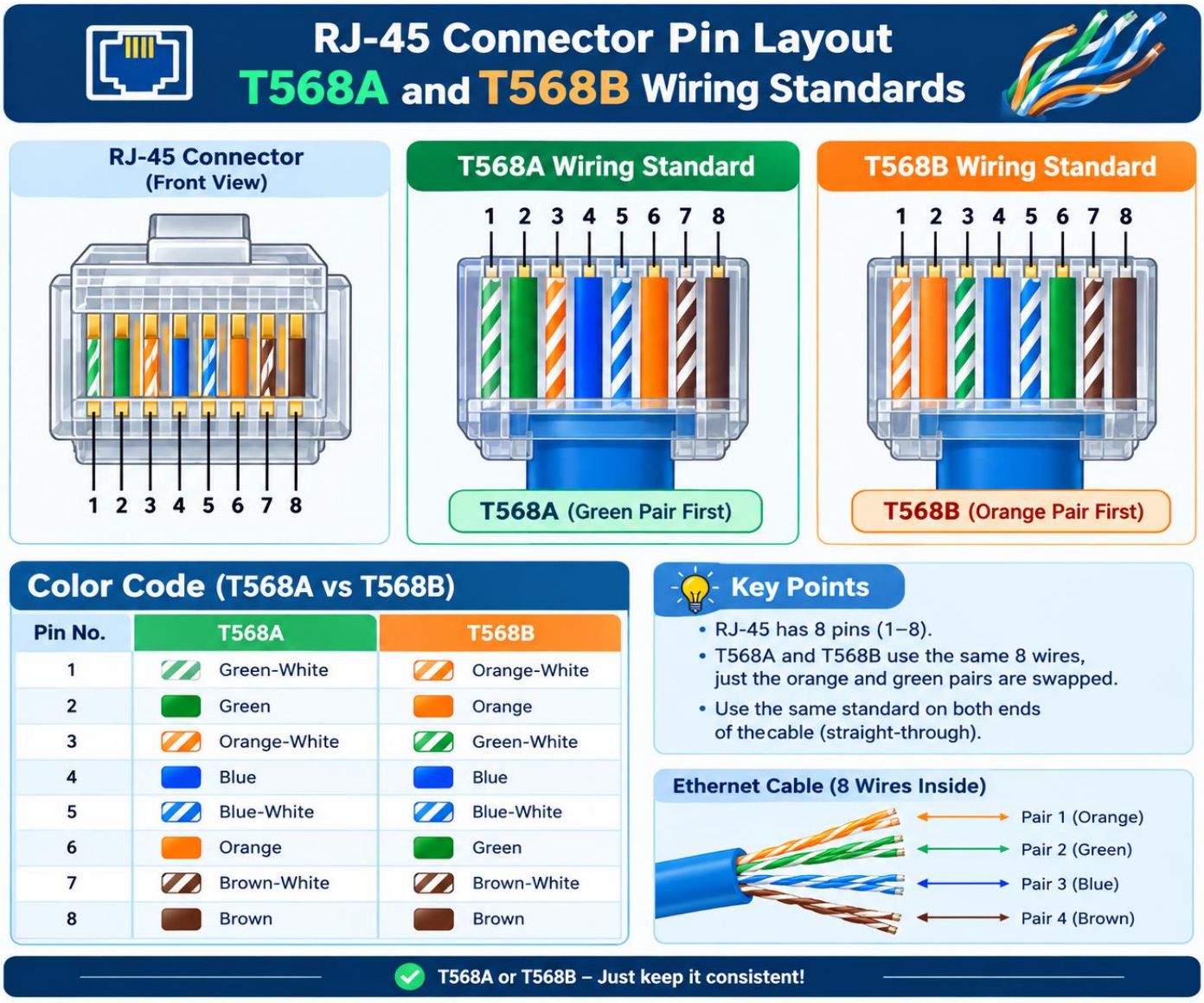
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Assignment-5

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1. Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.



2. Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

❖ Comparison of Ethernet Standards

Specification	10BASE-T	100BASE-TX	1000BASE-T
Speed	10 Mbps	100 Mbps	1000 Mbps (1 Gbps)
Cable Type	Cat3 or higher	Cat5 or higher	Cat5e or higher
Maximum Cable Length	100 meters	100 meters	100 meters
Connector Type	RJ-45	RJ-45	RJ-45
Common Use	Older computer networks	Fast Ethernet LANs	Modern LANs, gaming, and high-speed internet

3. Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why.Hint: Think about how transmit and receive pairs are wired in each cable type.

I would use a crossover cable to connect two laptops directly using Ethernet.

- A crossover cable swaps the transmit and receive wire pairs.
- This allows one laptop's transmit pair to connect to the other laptop's receive pair.
- A straight-through cable keeps the same wire order at both ends and is traditionally used to connect different devices, such as a laptop to a switch.

4. List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

Answer: Three Real-World Devices That Use Ethernet

No.	Device	Likely Ethernet Standard	Speed
1	Wi-Fi Router	1000BASE-T	1 Gbps
2	Desktop Computer (PC)	1000BASE-T	1 Gbps
3	Network Printer	100BASE-TX	100 Mbps

Explanation :

1. Wi-Fi Router: Connects to a PC or switch using an Ethernet cable.
2. Desktop PC: Uses Ethernet to connect to the router and access the Internet.
3. Network Printer: Uses Ethernet to receive print jobs from computers over a network.